

Question # 1

Q1: The ladder and the person weigh 15 kg and 80 kg, respectively. The center of mass of the 3.65 m ladder is at its midpoint (point C) as shown in Fig. 1. The angle $\alpha = 30^\circ$. Assume that the wall exerts a negligible friction force on the ladder at point A.

[20 marks]

- If $x = 1.2$ m, **determine** the magnitude of the friction force f_B exerted on the ladder by the floor?
- The person wants to climb to the top the ladder without slipping, **figure out** minimum coefficient of static friction between the ladder and the floor.

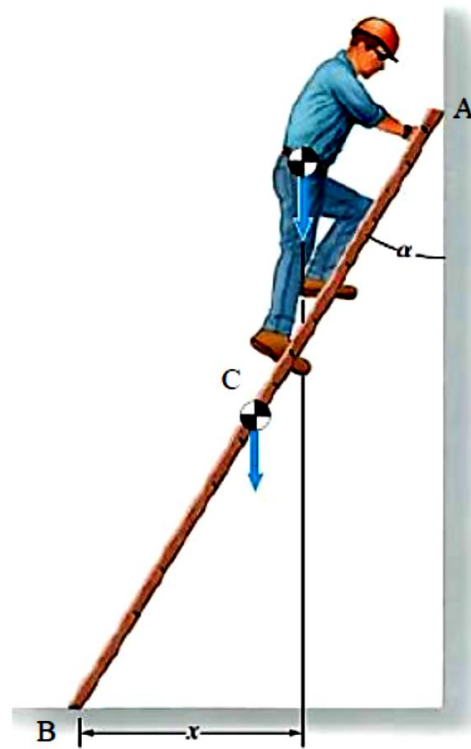


Figure 1. The person on the ladder

Ans: (a) $f_B = 340$ N, (b) $\mu_s = 0.536$

Question # 2

- The 700-N force is applied to the 100-kg block, which is stationary before the force is applied as shown in the Figure 2. Determine the magnitude and direction of the friction force F exerted by the horizontal surface on the block.

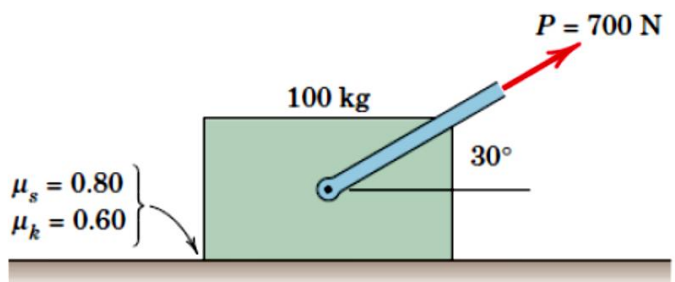


Figure 2. A block on a surface

Answer: $F_k = 378.6$ N (Direction is opposite to P_x)

Question # 3

Q2: The man exerts a force P on the car at an angle $\alpha = 20^\circ$. The 1760-kg car has front wheel drive (FWD). The driver spins the front wheels, and the coefficient of kinetic friction is $\mu_k = 0.02$. Snow behind the rear tires exerts a horizontal resisting force S . Getting the car to move requires overcoming a resisting force $S = 420 \text{ N}$. **Figure out** the force P , the man must exert.

[10 marks]

The center of gravity point is at 1.62 m from the rear tire as shown in Fig. 2. Remember to draw its FBD first.

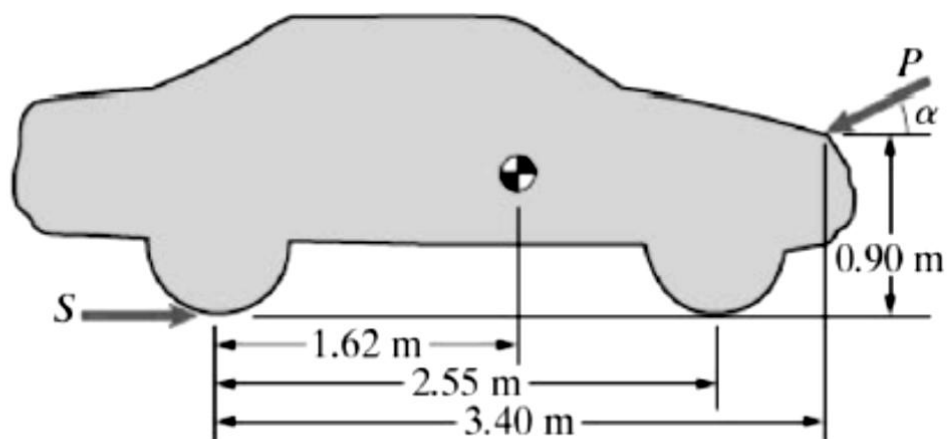


Figure 2. Car in Snow with Centre of Gravity point

Ans: $P = 684.214 \text{ N}$

Question # 4

- Q2** For the Figure 2 shown, if the coefficients of static friction at contact points *A* and *B* are $\mu_s = 0.4$ and $\mu'_s = 0.2$ respectively, **determine** the smallest force *P* that will cause the 150 kg spool to have impending motion.

[20]

CLO 02

Hint: FBD is given and assume that points A and B are at the verge of slipping.

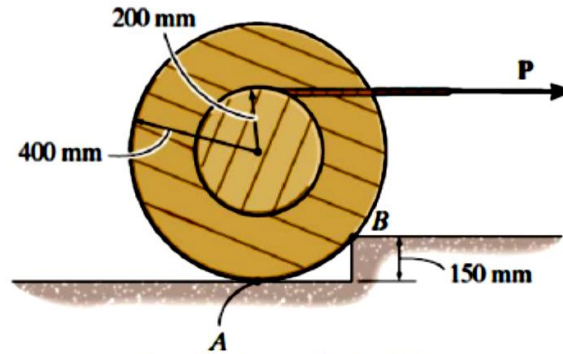


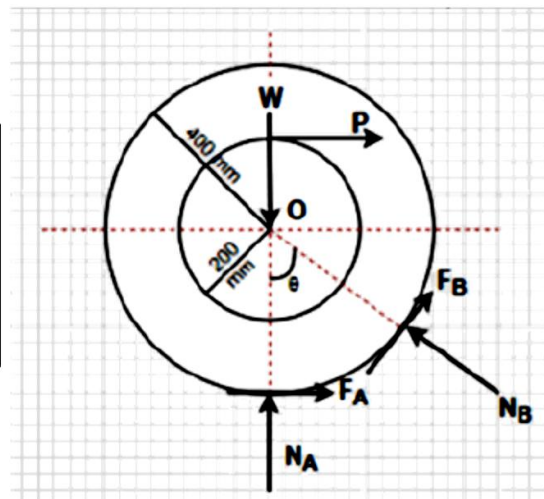
Figure 2a. Diagram for Dry Friction

Solution starts here ...

Finding the angle θ between N_A and N_B .

$$\theta = \cos^{-1} \left(\frac{400 \text{ mm} - 150 \text{ mm}}{400 \text{ mm}} \right)$$

$$\theta = 51.32^\circ$$



Answer: $P = 844 \text{ N}$

Question # 5

- Q2 (a)** **Determine** the range of values which the mass m_0 may have so that the 100-kg block shown in the Figure 2 will neither start moving up the plane nor slip down the plane. The coefficient of static friction for the contact surfaces is 0.35. [10 marks]
CLO 02

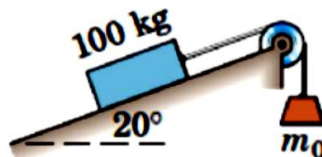


Figure 2. A 100-kg inclined block attached to a mass

Hint: There are two cases to be considered for the impending motion of the block and hence two FBDs.

Answers: Case-I: Up-- $m_0 = 62.4 \text{ kg}$, Case-II: Down-- $m_0 = 6.01 \text{ kg}$